Write-up

An analyst co-pilot that clarifies, checks definitions, picks the right data layer and shows its evidence.

01 What I built and why

1 Clarify — metric, period, population; asks if ambiguous

2 Verify definition — metric exists and its definition in registry + catalog

3 Route to a layer — Gold if it fits; Silver for detail; Bronze for raw audit

4 Query + evidence — read-only, bounded; one log per query, never overwritten

5 Test hypotheses — ≤3 per round, ≤3 rounds; stop when explained

6 Report — pattern ≠ cause; fixed PDF template, ≤10 pages

Read by every run, in Claude Code or Codex: AGENTS.md · SKILL.md · catalog · registry

Guardrails: read-only authorizer · row and time limits

Gold — 8 monthly tables · fast standard answers

Silver — 10 cleaned tables · detail for investigations

Bronze — 20 views over raw tables · audits

One SQLite database; layers built in one transaction with 122 reconciliation checks. Step 3 decides which layer each query reads.

- **Clarifies instead of assuming.** When the metric, period or meaning is open, it asks first and states every assumption it keeps.
- **Verifies the metric before computing it.** It confirms the metric exists and uses its approved definition and SQL from the metric registry, plus joins and known gaps from the data catalog. Unsupported requests are declined, not improvised.
- **Maps the question to a layer for speed.** Gold answers standard monthly questions directly; Silver keeps per-user detail; Bronze is only for auditing raw values. Fewer, smaller queries keep answers fast.
- **Evidence for every number and LLM-neutral by design.** Each query is logged with its SQL, layer and metric version. Definitions, rules and guardrails live in files, not prompts, so Claude Code and Codex behave the same and a lesson becomes a rule for every later run.

02 How it performed: 18 blind test questions, two rounds

	Q01	Q02	Q03	Q04	Q05	Q06	Q07	Q08	Q09	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18
Round 1	Pass	Fail (strict rule)	Pass	Pass	Pass	Pass	Fail (strict rule)	Pass	Pass	Pass	Partial	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Round 2	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Partial	Pass	Pass	Pass	Pass	Pass	Pass	Pass

Round 1: 15 pass · 1 partial · 2 fail **Round 2: 17 · 1 · 0**

Question type	Questions	Result	What happened
Simple numbers and filters	Q01 Q05 Q08 Q17 Q18	**All pass**	Exact to the user and the cent
Vague questions	Q03 Q10 Q15 Q18	**All pass**	Asked before choosing a metric, period or meaning
Unanswerable question	Q16	**Pass**	Declined and named the data that's missing
Data-quality traps	Q12 Q13 Q14	**All pass**	Caught both planted issues; didn't claim to have detected a reviewed one
Comparisons and lifecycle	Q04 Q06 Q09	**All pass**	Right time basis; groups reconcile to the total
Cohort conversion	Q07	**Fail → Pass**	Treated April, the start-up month, as a normal baseline; a new skill rule fixed it
Funnel drop-off	Q11	**Partial**	Right answer, but read April's start-up as a funnel decline; the rule didn't trigger
GM investigation	Q02	**Fail → Pass**	v1 right on numbers but too firm on some conclusions; after three certainty rules, v2 and a blind rerun both passed

Scored against values computed beforehand from the raw tables. Correct numbers alone don't pass: a silent choice of definition, a missing caveat or an overstated conclusion fails. Round 2: Q02 was revised after review and rerun blind; Q07 and Q11 were rerun, with Q01 and Q09 to check nothing broke; other results carry over. **Hardest:** comparisons that need context the data doesn't announce, such as a start-up month.

03 Running it for a 20-person product team

Time to answer (observed)	**Simple** (MAU by month): 2–3 min	**Normal** (refund impact): 5–6 min	**Complex** (MAU up, revenue flat): 10–12 min

- **How the team uses it:** analysts and PMs ask in plain English from Claude Code or Codex (later Slack or a web page), all on one shared repository, so definitions and guardrails are the same for every person and model.
- **Set-up:** build a data catalog and metric registry for the team's warehouse, with the analytics lead approving each definition, then the Gold and Silver layers the agent routes to.
- **Cost:** mostly model tokens; queries are nearly free. Not measured here: log tokens per question, then budget by question type. Answering from the smallest layer that fits keeps both cost and latency down.
- **Guardrails:** read-only access (database role plus the query tool's check), row and time limits, an evidence log behind every figure, no sensitive columns in the catalog, and a question back when the ask is unclear.
- **Freshness:** real data keeps arriving, unlike this dataset; the agent first checks a freshness tracker (last load per source and layer) and warns or stops when data is stale.
- **Quality over time:** a test set of questions with known answers (these 18, growing from real ones), rerun on every change to the skill, definitions or data; track pass rate, clarification rate, query errors and user ratings; turn failures into rules.

04 With more time

- Run the same questions in both runtimes for a direct Claude vs Codex comparison, plus held-out questions.
- Measure tokens and time per question to put a real cost on each question type.
- A helper that drafts the catalog and metric registry for a new database; a Slack or web front end.

Tools: Claude Code (Claude Opus 5.5, medium) built the framework, ran the GM investigation and the answer key; Codex (GPT Terra 6.1, medium) ran the evaluation; ChatGPT (Terra 5.6, later 6.1, medium) reviewed independently.